AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM340
Natural Language Processing

Final Examination
Summer Semester 2024/2025

Date: 03 / Sept. /2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed:** 180 minutes

Instructions:

- Total Marks: **100**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

CamScanner

PART 1

Multiple Choice Questions **(20 marks)**

Question 1, (20 marks, 2 marks for each question)
Answer the following MCQ's by selecting **the best** answer for each question. Write your **answer letter** in the answer sheet.

1) In a generic NLP pipeline, which stage involves converting pre-processed text into numerical forms that a machine learning model can understand?
   A) Data Collection
   B) Feature Engineering
   C) Model Evaluation
   D) Deployment

2) Which conversational agent category is designed to help users complete specific tasks, such as booking flights?
   A) Chatbots
   B) Corpus-based chatbots
   C) Task-based dialogue agents
   D) Neural chatbots

3) When processing a Boolean query like (A OR B) AND (C OR D), which of the following is an effective query optimization strategy?
   A) Process the terms in alphabetical order.
   B) Always process the AND operation first.
   C) Estimate the size of each OR clause and process the one with the smaller estimated size first.
   D) Convert the entire query into a single OR expression.

4) In the Vector Space Model, why is the angle between document vectors preferred over Euclidean distance as a measure of similarity?
   A) The angle is easier to compute for sparse vectors.
   B) The angle is not affected by vector length, whereas Euclidean distance is.
   C) Euclidean distance can only be used for binary vectors.
   D) The angle is always a positive value, which is required for ranking.

5) Which of the following word categories is considered "open class" because new words are frequently added to it?
   A) Determiners
   B) Pronouns
   C) Conjunctions
   D) Nouns

6) The Word2vec model learns dense word embeddings through self-supervision. What is the binary prediction task it is trained on?
   A) Predicting whether a sentence is grammatically correct.
   B) Predicting whether a word is a noun or a verb.
   C) Predicting whether a candidate word is likely to appear near a target word.
   D) Predicting the sentiment of a document.

CamScanner

7) In the tf-idf weighting scheme, what is the effect of the idf component?
   A) It increases the weight of terms that are frequent within a single document.
   B) It gives a higher weight to terms that are rare in the overall collection.
   C) It normalizes the term frequency to be between 0 and 1.
   D) It removes stop words from the document vectors.

8) In a feedforward neural language model that uses a fixed-size window of N-1 words to predict the Nth word, what is a primary limitation?
   A) It cannot learn word embeddings.
   B) The model size grows exponentially with the vocabulary size.
   C) The fixed window size limits the amount of context the model can use.
   D) It can only be used for text classification, not language modeling.

9) In task-based dialogue agents, what is the primary purpose of the GUS architecture?
   A) To engage in open-ended, human-like chatting for entertainment.
   B) To generate responses by retrieving them from a large corpus of conversations.
   C) To use a frame with slots to gather the necessary information to complete a specific task.
   D) To learn from user feedback using reinforcement learning.

10) In log-frequency weighting, why do we use the logarithm of term frequency instead of the raw count?
   A) To reduce the impact of very high term frequencies
   B) To increase the effect of rare terms
   C) To convert frequency to binary values
   D) To simplify calculation of inverse document frequency

PART 2

Short Essay Questions **(20 marks)**

Question 2, (20 marks)
Answer the following questions:

a) Why is an inverted index a more efficient data structure than a term-document incidence matrix for large document collections? (5 marks)

b) How do the evaluation methods for a chatbot (like ELIZA) differ from those for a task-based dialogue agent (like GUS)? (5 marks)

c) In vector semantics, what is meant by a word's connotation? List the three affective dimensions used to measure it? (5 marks)

d) Despite their advantages, RNNs suffer from a key problem that can make learning such long-distance dependencies difficult in practice. Name and briefly describe this problem. (5 marks)

CamScanner

PART 3

Short problems/questions **(60 marks)**

Question 3, (20 marks)

a) You are given the following inverted index postings lists for three terms:
brutus: [2, 4, 8, 16, 32, 64, 128]
caesar: [1, 2, 3, 5, 8, 13, 21, 34]
calpurnia: [2, 13, 16]

A user submits the following Boolean query (Q):
(brutus OR caesar) AND NOT calpurnia

What is the output of applying the merge algorithm to process the query Q, show the required steps and the resulting list of docIDs? (10 marks)

b) Given the following document frequencies (df) for a small collection of terms:

Term	*df*
nlp	150
retrieval	300
model	250
system	400

A user submits the following query (Q):
(retrieval OR system) AND (nlp OR model) AND (model OR system)

What will be the optimized query processing order for Q (**show the required steps**)? (10 marks)

Question 4, (20 marks)

Given a document collection with N=1,000,000 documents. You need to rank three documents (D_1, D_2, D_3) for the query Q = "**new york times**". The raw term frequencies (tf) and document frequencies (df) are provided in the table below:

Term	*df*	*tf in* D_1	*tf in* D_2	*tf in* D_3
new	50,000	15	0	5
york	40,000	12	0	5
times	80,000	5	10	0

Answer the following (**show your calculations for each step**):

a) Calculate the inverse document frequency (idf), and the log-frequency weighting (tf) values for each term? (6 marks)

b) Calculate the *tf-idf* for each term, then state decide which term has the highest *tf-idf* value in D_1? (4 marks)

CamScanner

c) Calculate the length of each *tf-idf* vector for Q, D_1, D_2, and D_3? (5 marks)

d) Calculate $cos(Q, D_1)$, $cos(Q, D_2)$, and $cos(Q, D3)$ and state which document is more relevant to the query? (5 marks)

Question 5, (20 marks)
Consider the following paragraph:

> *The new tablet's design is brilliant and the initial setup was surprisingly easy. However, the battery life is truly awful, and I found the customer service to be unhelpful. I would not recommend this product.*

And given the following lexicons:

> **Positive:** brilliant, easy, recommend, great
> **Negative**: awful, unhelpful, bad, poor

Use a simple neural network with one layer to decide if a paragraph expresses a **positive** (y = 1) or **negative** (y = 0) sentiment. Use the **sigmoid** as an activation function and Suppose: **w = [0.6, -0.8, -1.5, -1.0]**, **b = 0.1**, and the **features** will be:

- x_1: The number of positive words in the paragraph.
- x_2: The number of negative words in the paragraph.
- x_3: A binary feature that is 1 if the word "not" appears, and 0 otherwise.
- x_4: A binary feature that is 1 if the count of negative words (x2) is greater than the count of positive words (x1), and 0 otherwise.

Answer the following (**show your calculations for each step**):

a) Determine the values for the feature vector [x_1, x_2, x_3, x_4] based on the provided paragraph? (4 marks)

b) Use the feature vector and network parameters to compute the value of z and the final activation y. (7 marks)

c) Decide whether the neural network classifies the paragraph as positive or negative. (Assume a threshold of 0.5)? (3 marks)

d) The feature set for this model is based on a "bag of words" approach. Consider the sentence: "The performance is **not bad**, it's actually **brilliant**." Explain why this model would likely misclassify this sentence and suggest one specific way the features could be improved to better handle such cases? (6 marks)

End of questions

CamScanner